Time-Resolved Fluorescence Immunoanalyzer

LTRIC-600

Operator's Manual

Revision 1

Manual version	Revision date	Changes
Version 1	2015.07.01	New document

For supervisor and operator, please read this document BEFORE your operation!

Reference of WARNING and ATTENTION are listed as follow:

⚠ WARNING: Inappropriate operation might lead to SERIOUS INJURY or DEATH.

⚠ ATTENTION: Inappropriate operation might lead to INJURY or FINANCIAL LOSES.

During the Installation:

⚠ ATTENTION

Select appropriate installation location:

Please start the installation of LTRIC-600 in an appropriate location after reading the relevant sections of this manual.

Use ONLY the AC/DC power adapter and power cable provided with LTRIC-600. Do NOT use the same electrical socket with other electrical appliances.

May cause FIRE or ELECTRIC SHOCK.

Make sure the fork is clean and plugged into the electrical socket tightly.

Please disconnect and reconnect the power cable several times per year.

May cause FIRE or ELECTRIC SHOCK when the fork is dusty or not tightly connected.

During the operation

⚠ Notice, operation will not be allowed till you understand all the precautions.

PROHIBITED, warning of inappropriate operations.

⚠ WARNING

Please prevent with cautions from infection.

LTRIC-600 must be operated by skilled persons with the training to perform clinical examination and handle medical wastes.

Biological samples might be contaminated by infectious pathogens. Mishandling of biological sample may cause pathogen infections to the operator or other individuals. Please wear appropriate PPE (eye protectors, gloves, medical masks etc.) when performing clinical examination or doing the maintenance work of this instrument.

All used test strips, tubes, tips and reagents are biohazards. Please be equipped with appropriate PPE(eye protectors, gloves, medical masks etc.) to operate the system and analyze samples, and make appropriate pre-preparation according to the instructions on the next sections and the requirement of relevant laws and policies.

⚠ ATTENTION

Do NOT operate LTRIC-600 without instruction in this manual. Otherwise might cause failure, accident or inaccuracy of measurement, etc.

Please shutdown LTRIC-600 and disconnect power cord IMMEDIATELY if something abnormal (burnt smell, for instance) occurs, and then contact our service center.

May cause fire or electric shock if keep running under an abnormal condition.

Please DO NOT put your finger or any other rod-shaped items into the test strip entrance of LTRIC-600.

There is an electrical-driven motor inside LTRIC-600. Severe injury might occur because of getting stuck!

Please do not turn on/off the instrument by directly connect/disconnect power supply.

Otherwise might cause fire, electric shock or damage to PCB inside the device, etc.

Use power switch at the left side of LTRIC-600.

Do NOT damage the power cable.

Over stretching, forcibly bending might damage the power cable, and result in fire or electric shock.

Do NOT take out the fork from the socket by pull the power cable.

⚠ ATTENTION

Wear personal protective equipments (PPEs) when performing maintenance work.

May get injured or infected without fully understanding of relevant knowledge during the maintenance work. Person in charge of LTRIC-600must be well trained and fully understanding of relevant knowledge. All the operations must be done according to this manual.

If you have any questions about maintenance work, please contact our service center.

Please dispose the wastes appropriately.

Medical wastes like used samples, test strips and tips must be classified according to their characteristic and disposed appropriately. Please be equipped with infection-proof supplies during all the procedure. Do NOT contact the wastes directly by hand.

Please make the appropriate pre-disposal according to the indications on the next sections and the requirement of relevant laws and policies about medical waste, to minimize any potential environmental contamination.

Please be equipped with personal protective equipments.

Please make sure you are equipped with infection-proof supplies (including special eye protectors, gloves, respirators etc.) when handling samples or medical wastes, and calibrating the instrument with reference standards.

Place test strips only where safety requirement of the instrument manual is met.

If the liquid leaks into the instrument, it may cause short circuit or decrease of insulation ability, even electric shock.

Other notices

Warning Signs

Warning signs are attached to this instrument. Please confirm the meanings of these signs and use this instrument safely.

A: Pollution, Warning

May get infected by samples when using the kit. Please be equipped with infection-proof PPEs, such as gloves during the operation. Avoid direct contact or you might get infected.

B: Running Machine, Attention

Do NOT put your fingers into test strip entrance when the instrument is running, otherwise you may get injured.

C:In Vitro Diagnostic Medical Device

Time-Resolved Fluorescence Immunoassay Analyzer LTRIC-600 is an in vitro diagnostic device.

Please contact our service center if the warning or attention signs are stained, fallen off or unclear. We will deliver new signs for replacement.

Please keep this manual ready for reference. Besides, please transfer this manual to the successor when person in charge of this equipment is replaced.

Please contact us for repair.

Disassembling, repairing and rebuilding by yourself may cause fire or electric shock.

Address of our service center is listed at the end of this manual.

Lumigenex (Suzhou) Co.,Ltd.

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1. Introduction

First, we'd like to thank you for purchasing LTRIC-600. This is the English version of manual of Time-Resolved Fluorescence Immunoanalyzer LTRIC-600 including the constitution, operation approach and other things related to this instrument.

Time-Resolved Fluorescence ImmunoanalyzerLTRIC-600 is a delicate device, please read this instruction manual carefully before using it. If the operator has not installed, stored or operated this instrument according to this instruction manual, it may cause damage to the device, injury to the people and other medical accidents!

Time-Resolved Fluorescence Immunoanalyzer

LTRIC-600

Summary

The Time-Resolved Fluorescence Immunoanalyzer LTRIC-600 is an instrument for quantitative evaluation of immunoassays using the time-resolved fluorescence technique. The instrument is for ***in-vitro diagnostic*** use only. It should be used conjunctly with the quantitative TEST KITs produced by our company. Please refer to

the package inserts accompanying the TEST KITs for detailed information on specific tests.

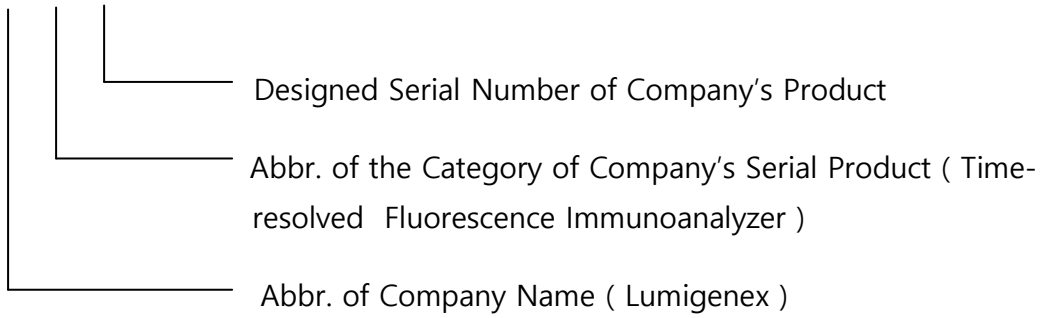
Time-Resolved Fluorescence Immunoanalyzer LTRIC-600 is basically consists of shell, photo electric sensor, barcode scanner and its drive motor system, signal processing system, display device, printer, USB port, IC card socket and power adapter.

Applicable scope of this product: Time-Resolved Fluorescence ImmunoanalyzerLTRIC-600 is an in vitro diagnostic product. It should be used conjunctly with the quantitative TEST KITs produced by our company.

This instrument applies only to the test of the kit with Time-Resolved Fluorescent Immunochromatographic Assay Technology, and be used conjunctly with the quantitative detection kit produced by our company.

Analyzer Version: LTRIC-600

LTRIC-600



Overvoltage Category: Category II; Pollution Level: Level 2

Test Principle

The test strip contains two monoclonal antibodies against target antigen products in test sample. One of the antibodies is coupled with nano-particles which embedded with fluorescence probe with time-resolved specificity. The antibodies form a sandwich complex with the target antigen molecular in the

test sample. The antigen molecular in test sample reacts with the first antibody-nanoparticles in sample pad then passes through the detection zone in which the sandwich complexes accumulate (test line) . Excess fluorescence microsphere labeled antibodies accumulates along the control line. The intensity of the signal line increases in proportion to the target molecular concentration. The signals in test line and control line are captured by LTRIC-600 reader respectively. The optical system of the LTRIC-600 reader instrument detects the two lines and measures the intensity of the signal line. The integrated software converts the signal intensity to a quantitative result and shows it in the display.

Content of the Pack

Product can be installed by the user after delivering. Please install this instrument according to the indications from this instruction manual. Please check carefully to make sure all accessary parts included and contact us for any missing parts.

Content List:

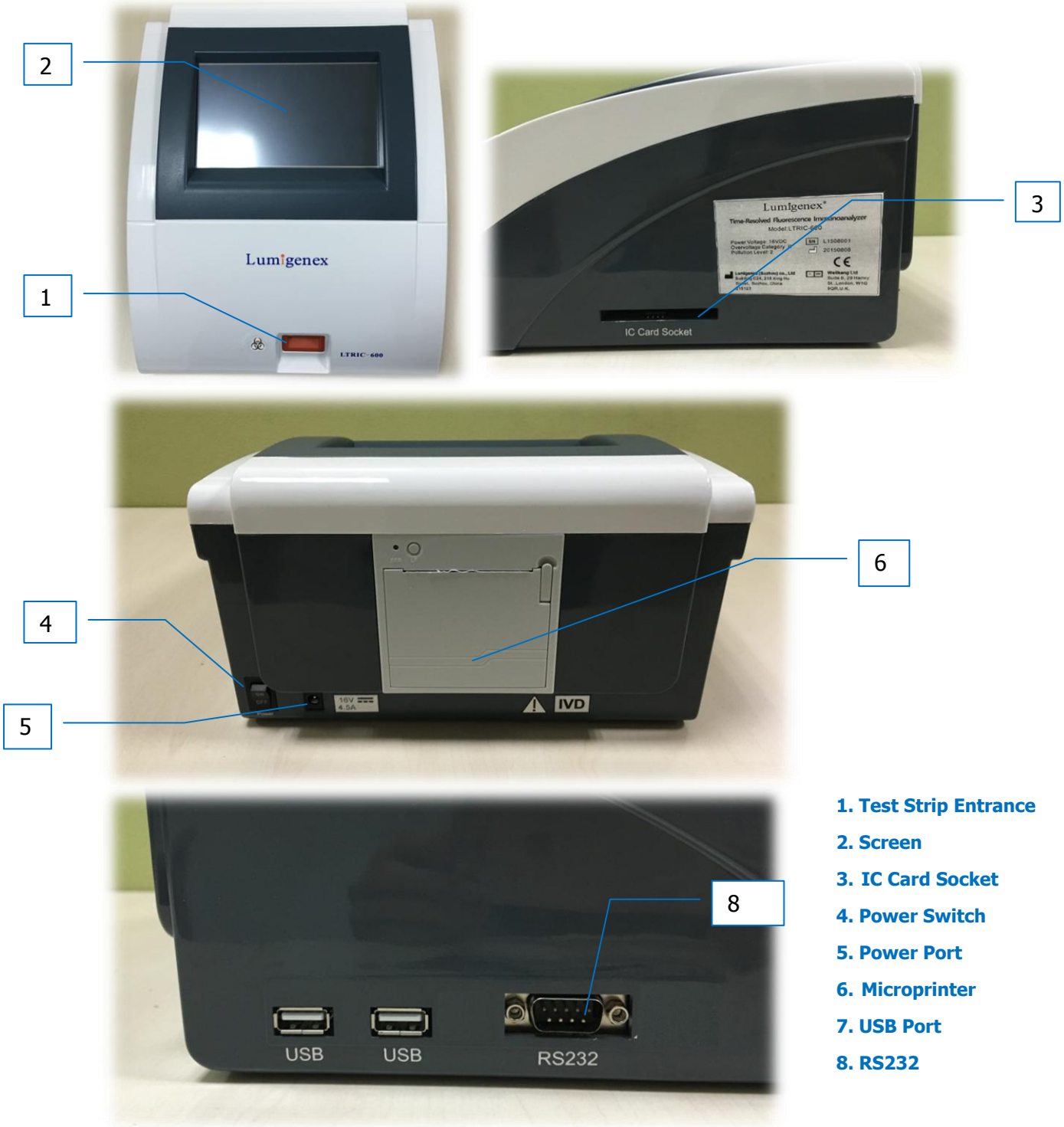
LTRIC-600	1
Power Adapter	1
Operator's Manual	1

2. The Immunoanalyzer and its accessories

Immunoanalyzer

The appearance of LTRIC-600 has been shown at Picture 2-1 & Picture 2-2.

Picture 2-1 Appearance of LTRIC-600



1. Test Strip Entrance
2. Screen
3. IC Card Socket
4. Power Switch
5. Power Port
6. Microprinter
7. USB Port
8. RS232

Front

Back

Left Side

Right Side

Test Strip Entrance

Test Strip Entrance is used for inserting test strip.

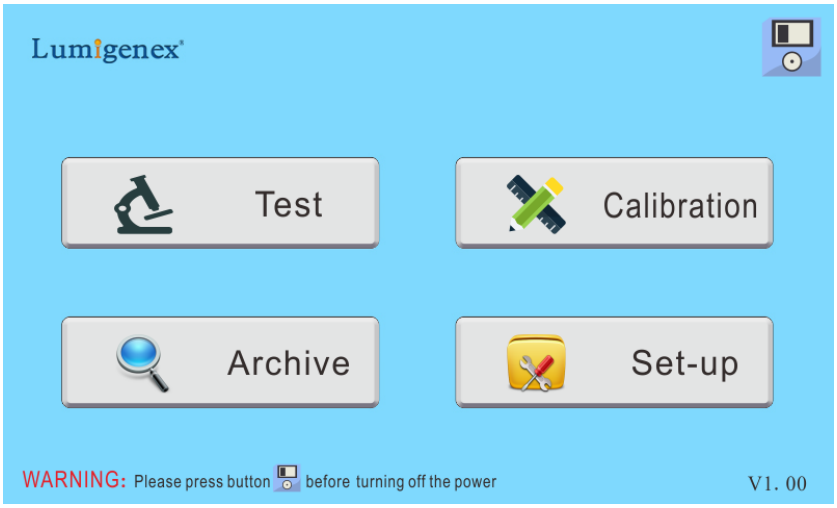
Picture 2-3 Test Strip Entrance

Keep the test side UPWARD while inserting.

Screen

LTRIC-600 uses a LCD touch screen for operating and displaying.

Testing results and other information will be displayed on the screen.



IC Card Socket

IC Card Socket is used for reading IC card information.

Picture 2-4 IC Card Socket

Keep the chip DOWNWARD while inserting.

Power Switch

Power Switch is used to turn on or shut down the instrument.

Picture 2-5 Power Switch

'OFF' refers to shut down,'ON' refers to connected.

Power Port

Power port is used for connecting with the output port of the power adapter.

Current through this port has to be 16V, 72VA, DC.

Port size is 5.5*2.1mm.

Picture 2-6 Power Port (Disconnected)

Picture 2-7 Power Port (Connected)

Microprinter

Microprinter is used for printing out testing results.

55mm thermosensitive print paper is applied to this printer.

Picture 2-8 Microprinter

USB Port

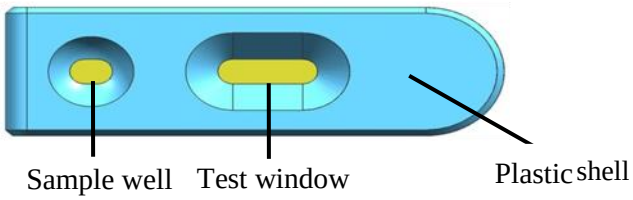
Picture 2-9 USB Port

RS232

RS232 served as data communication port, it can connect with LIS system.

Picture 2-10 RS232

Test strip



3. Installation

LTRIC-600 is a delicate analytical instrument. In order to work stably, it demands correct installation and adjustment. Please install and store this instrument according to the instruction manual when delivering, installing or shifting.

Transport and Storage Condition

Transport Temperature: -10℃ ~ 55℃

Transport Humidity: <90%

Storage Temperature: -10℃ ~ 55℃

Storage Humidity: 20% ~ 80%

Please store indoor.

Working Condition

Please install LTRIC-600 on a plat table without harmful gas, dust, vibration, direct sunlight or wind。Please use Time-Resolved Fluorescence Immunoanalyzer LTRIC-600 under the condition as below :

Temperature	**5℃ ~ 40℃**
Humidity	**<90%**
Power Supply/ Consume	**100~240VAC , 72VA , 50/60Hz**
Overvoltage	**Category II**
Pollution Level	**Level 2**
Dust	**Correspond to a Normal Office**
Top Height Above Sea Level	**2000m**

Please do not use this instrument under an extreme climate condition. Extreme temperature difference would cause condensation, and increase the risk of electric leakage and disable the instrument.

LTRIC-600 is designed according to EN /IEC 61010-1:2010, IEC 61010-2-101:2015 ; EN 61326-1:2006, EN 61326-2-6:2006.

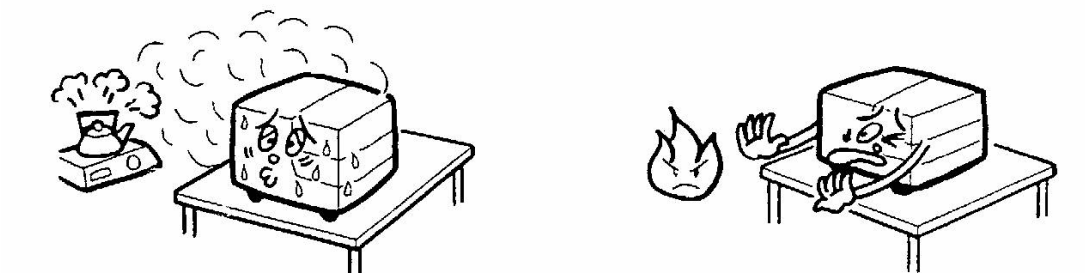
It may cause Radio Interference at home.

DO NOT use LTRIC-600 near a strong radiation source (an unblocked radio source for instance). It may cause Functional Faults.

Installation Location

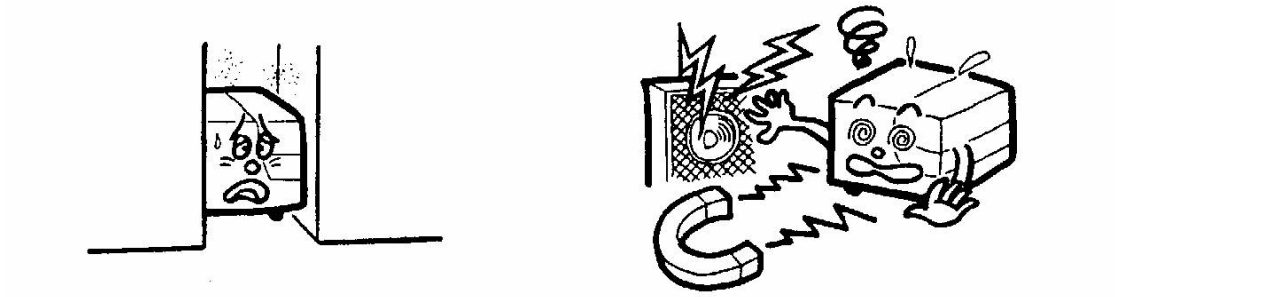
Please AVOID locations as followed:

WithHigh HumidityNear Naked Fire

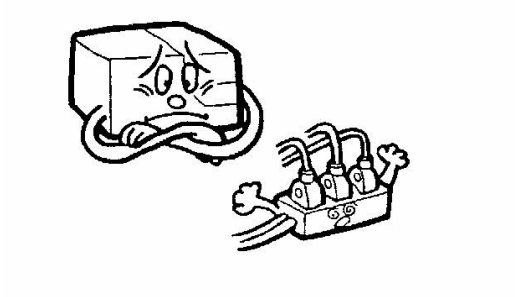


·Narrow and Unventilitive·

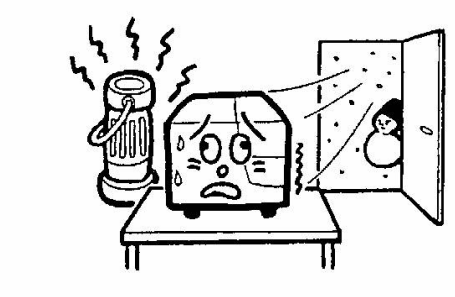
Exposed to High Frequency Electromagnetic Wave



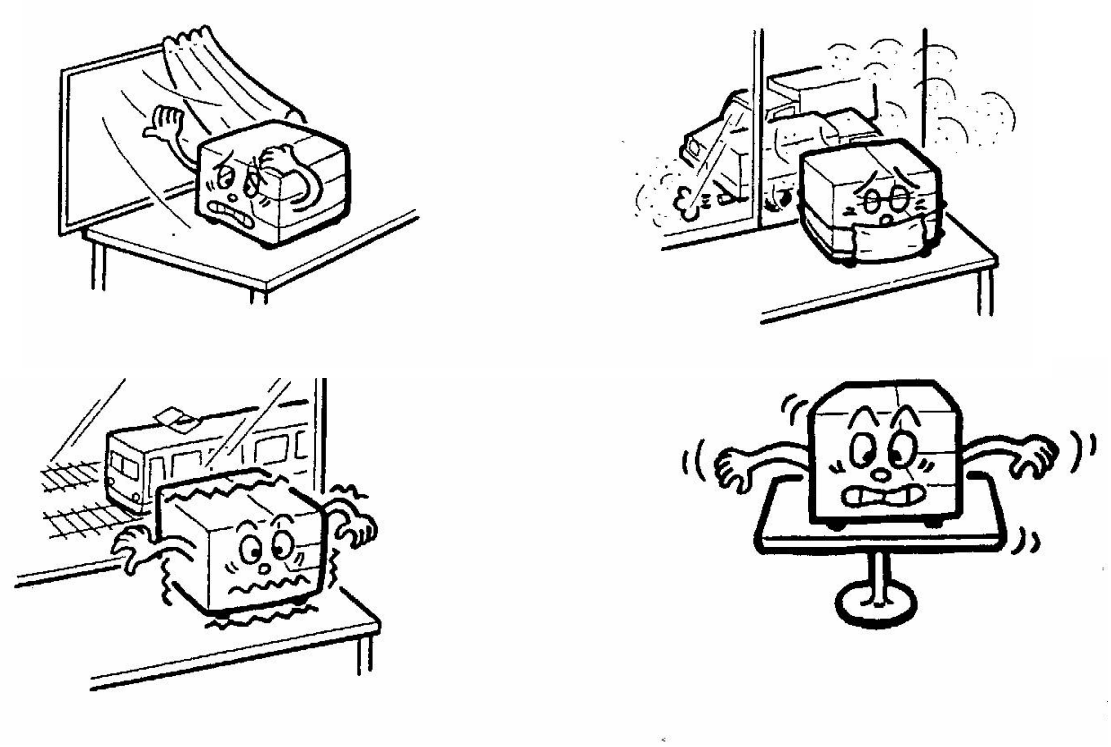
·With Massive Power Voltage Change·



With Acute Temperature Change



·Windy·Dusty



Installation Space

Dimension of LTRIC-600 is 255mm (Width)×315mm (Length)×150mm (Height)

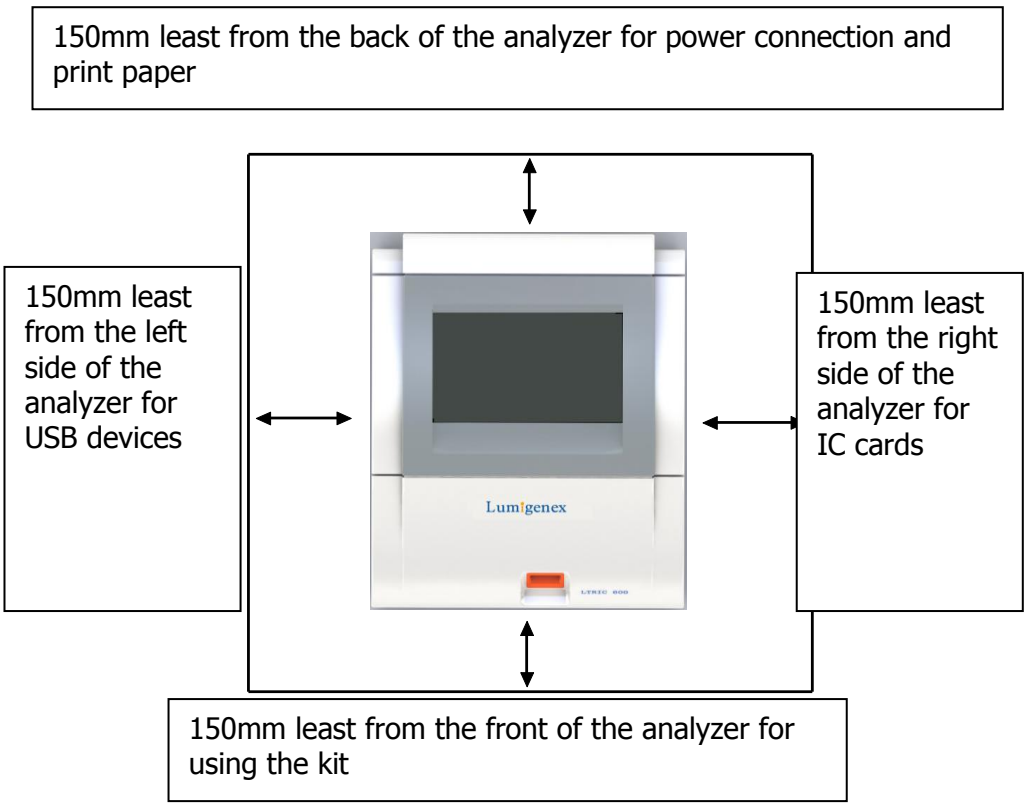
Space reserved for installation requires:

150mm least from the left side of the analyzer for USB devices

150mm least from the right side of the analyzer for IC cards

150mm least from the back of the analyzer for power connection and print paper

150mm least from the front of the analyzer for using the kit



Picture 3-1 Power Port

Connection between power adapter and LTRIC-600 has been showed in picture 3-2

Picture 3-2 Power adapter connection

Power Supply

①Ensure your power supply fits the parameters below:

LTRIC-600 connects with the power supply through a power adapter, and ON/OFF the current with a power switch. The Adapter Parameters are followed:

Input: AC100V-240V , 50/60Hz.

Host Input:DC16V , 72VA

Make sure the switch on the back turned to OFF.

Connect the output port of the adapter with the power port on the instrument.

② Insert the adapter's plug into a 100~240V power socket after the power port has been connected correctly.

DO NOT use the same power socket for LTRIC-600 and High-Power Electrical Appliance such as fridges and compressors.

③Turn on/off LRTIC-600 with the power switch after the adapter has been connected correctly.

Picture 3-5 Power Switch

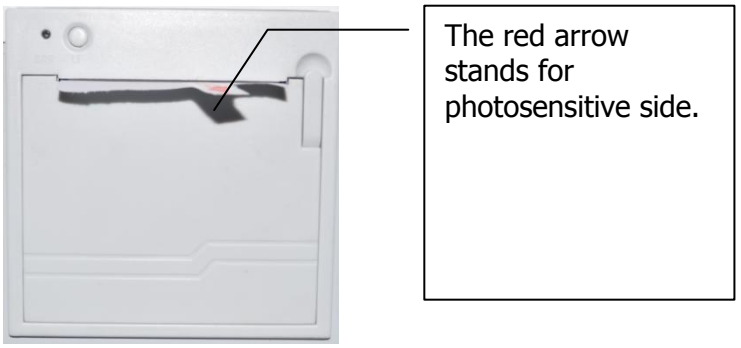
OFF refers to disconnected, ON refers to connected.

Print Paper Installation

First, press down the switch on the printer

Next, open the box and put the paper in

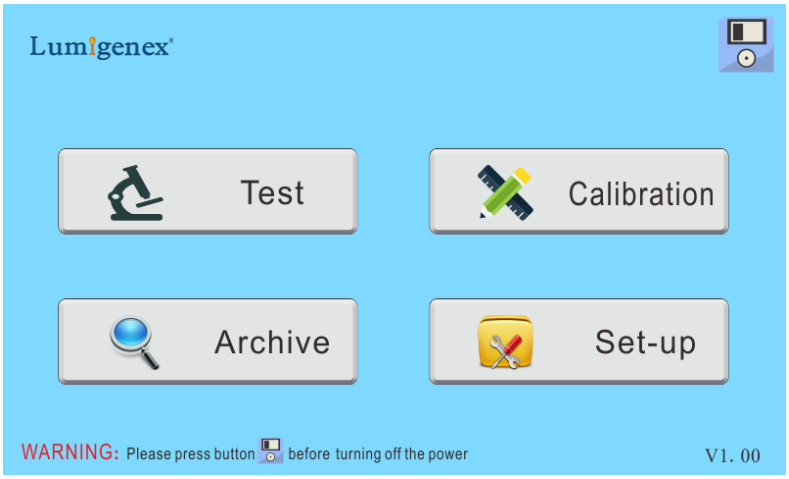
Finally, close the cover and finish the installation.



4. **Immunoanalyzer Setup**

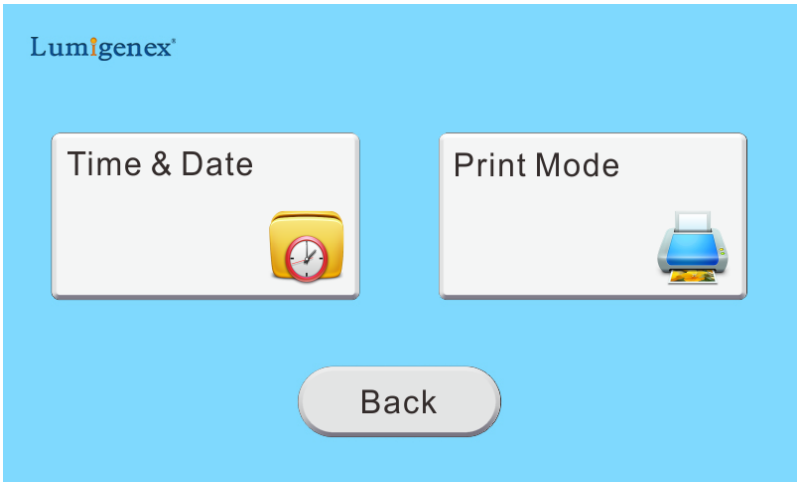
System has been set up already. You can skip this step to run the test. But if you need to change the settings, you can process with following steps:

Enter the Main Interface:

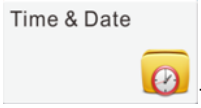


Click

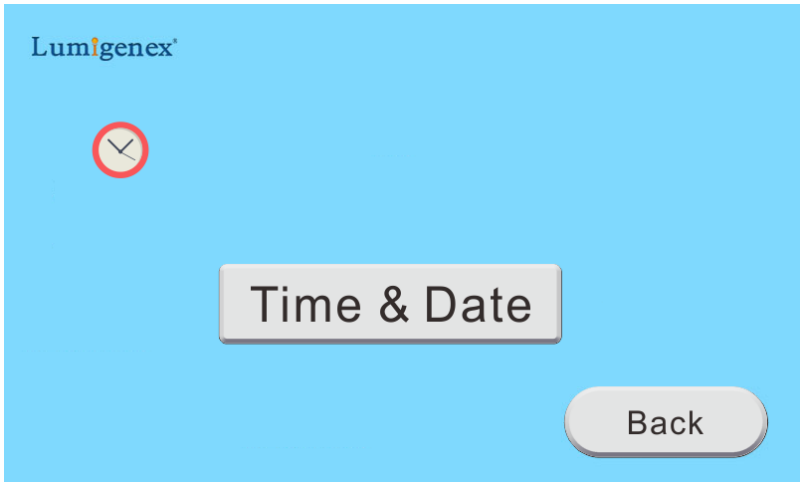
to enter Settings Interface as below:



Setting the date & time

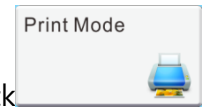
Click 

to enter Time & Date Settings Interface.

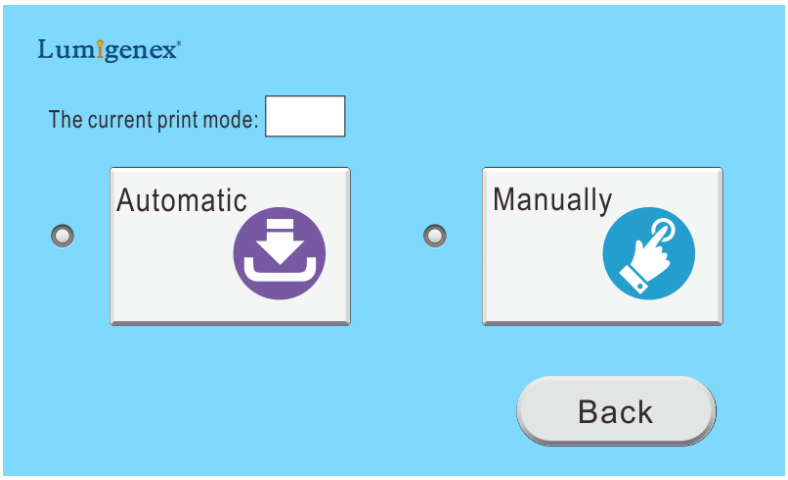


Click Time & Date to type your time & date information. Click Enter after you finished.

Setting the print mode

Click 

to enter Print Settings Interface



Click Automatic or Manually to switch print mode. Default mode is Manually. If you switch to Automatic, test result will be print out automatically every time you finish a test.

5 . Preparing to Test

Sample material

Following items should be ready before the test:

Lumigenex Test Kit for the desired test with the supplied IC Card

Sample

Prepare the Kit

Read the TEST KIT instruction carefully. Make sure your operation appropriate.

Wait for the test kit adapting to ambient temperature before running the assay.

Take out the strip and place it on a clean platform.

Add the sample into the well on the strip.

Test strip should be used within 1 hour after being taken out.

Warnings and Precautions:

- **This is a single-use disposable test kit for *in vitro* diagnostic use only.**
- **The validity of test kit sealed in aluminum foil pouch is 12 months under condition of 2 °C -8°C. Do not use the kit after the expiration date.**
- **Keep the test kits in the sealed pouch until ready for immediate use. The unsealed kit is stable within 1 hour under ambient condition.**
- **The test kits, sample buffer and specimens should be balanced to room temperature before use. Optimal test condition is at the temperature of 15 °C -30°C, 40%-60% humidity.**
- **DO NOT perform the test if any part of the test kit is broken.**
- **DO NOT use test kits with unmatched lot number IC card, or interchange different lot number kits in one set of tests.**
- **Each transfer pipette should be used for one patient. Dispose used pipettes as infectious biological waste. Proper handling and disposal methods should be established by the laboratory in accordance with local, state and federal regulations.**
- **DO NOT take any decision of medical relevance once using the test kit without your doctor's advice.**

Prepare the Sample

Separation should be done as soon as possible after sample collection to avoid hemolysis.

Add sample vertically into the sample receiving well on the test strip.

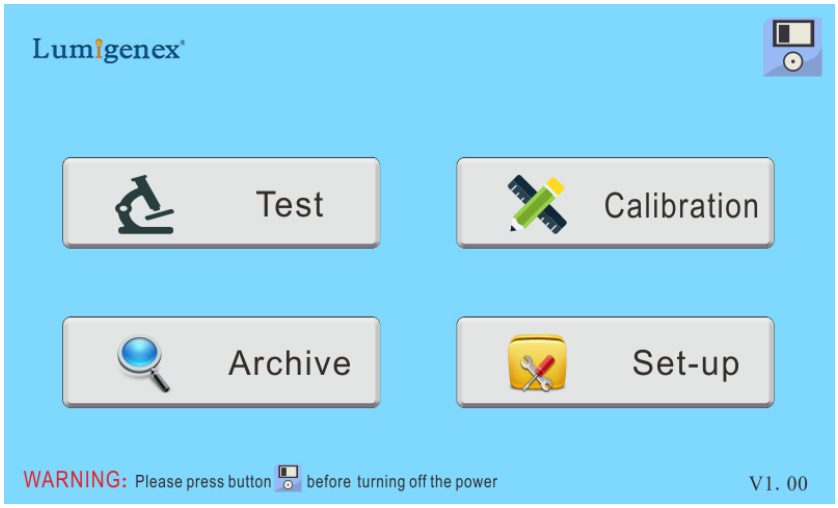
Wait for several minutes as required.

Insert the test strip into the test strip entrance on LTRIC-600 and run the test.

Test result should be taken within 1 minute so to remain accuracy.

Used test strip and sample should be disposed appropriately.

⚠ **Follow the information on correct handing of the TEST KITs in the package insert.**



Powering on the Immunoanalyzer

Before Starting:

Confirm that all current supply devices are connected correctly.

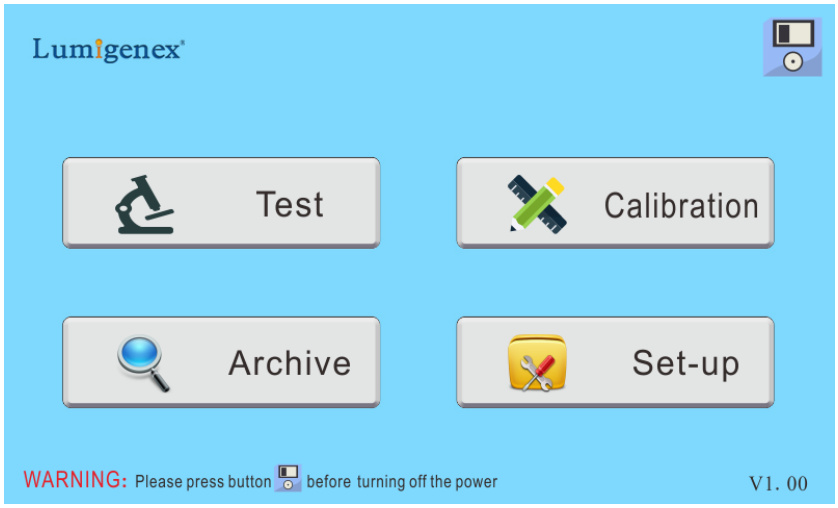
① Turn on the power switch to start LTRIC-600.

Picture4—1 power switch

'OFF' refers to turn off, 'ON' refers to turn on

② 'Initializing' will be displayed on the screen after starting.

Then follow into main interface as below:



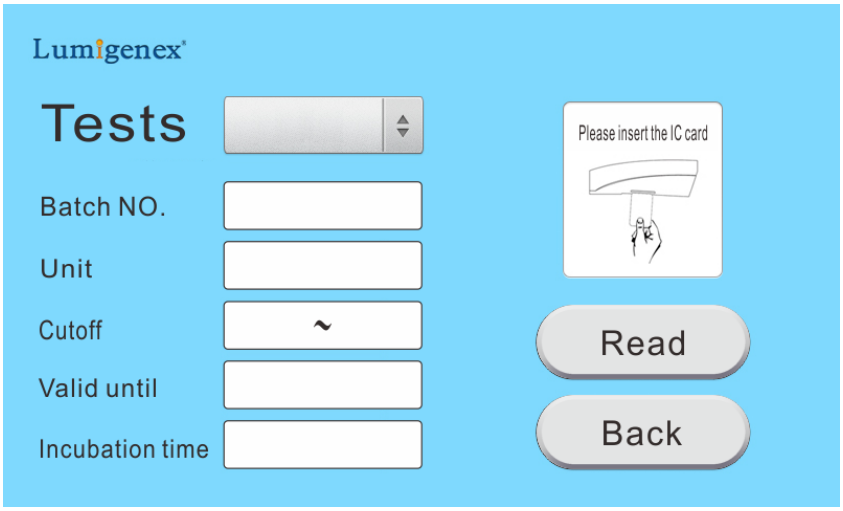
Keep LTRIC-600 on for 30 minutes before testing could stabilize its performance.

Code card (IC card)

Calibration has to be done before testing if it is your first time using LTRIC-600 or batch number of the test kitis changed.

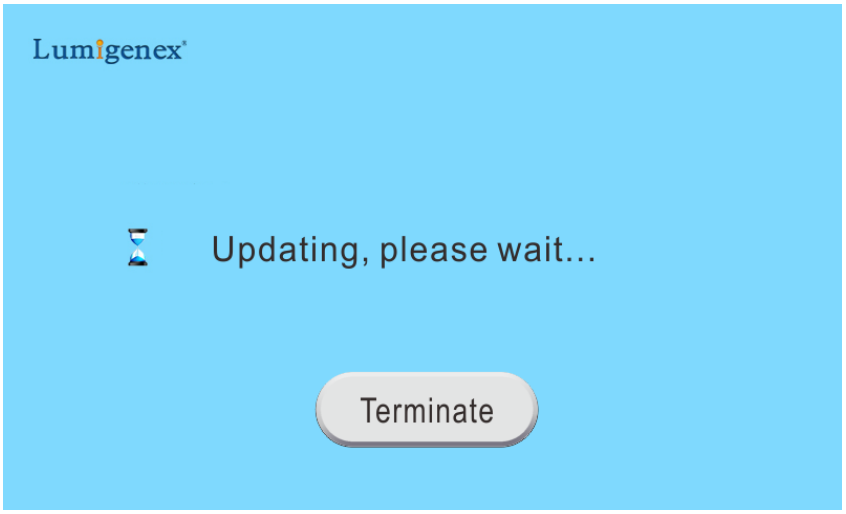
Click

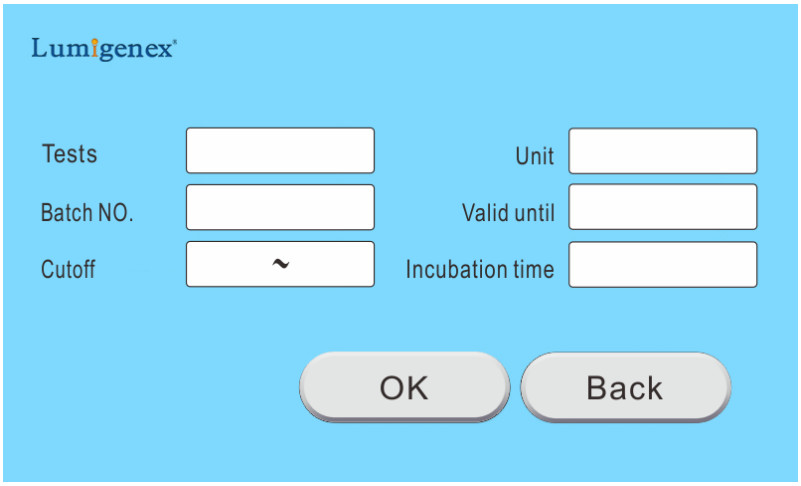
on the Main Interface to enter Calibration Interface.

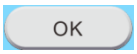


There is an IC card socket on the right side of the instrument. Insert the IC card with the chip downward, click

to upload the IC card information.

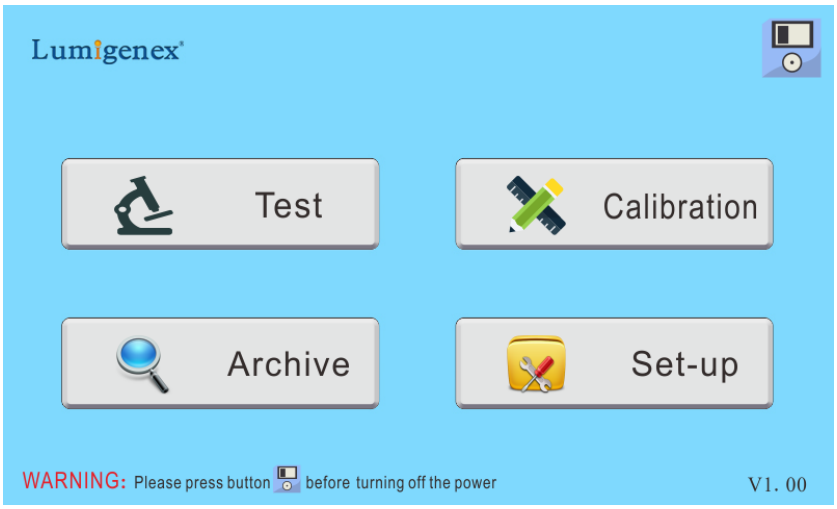




Click 

after the updating finished. Click BACK to return to main interface.

Logging in



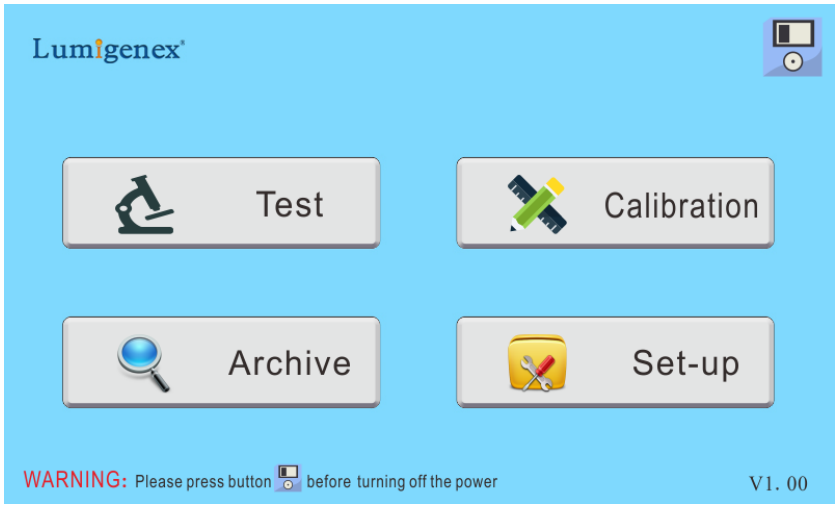
This is the Main Interface.

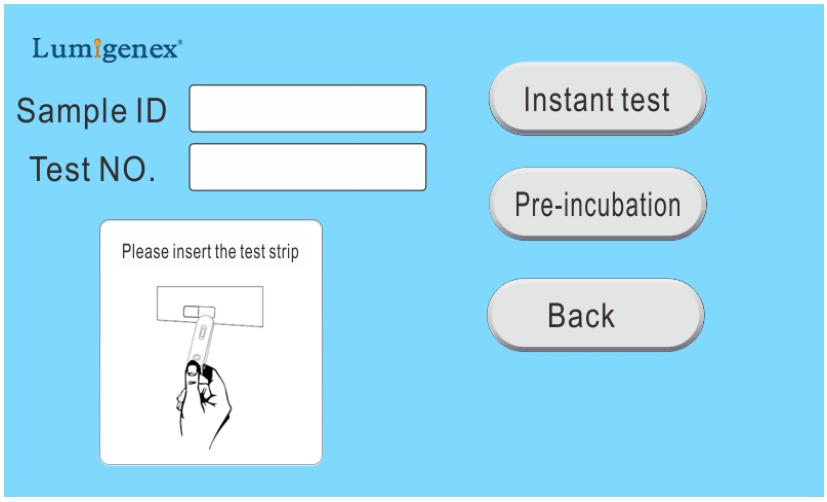
6. Performing a Test

Inserting the test strip

Click

on the main interface to enter Testing Interface.



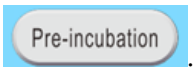


Default sample identifier is blank.This identifier can be formed by letters and numbers. You can build the identifier with an external bar code scanning gun or an external keyboard or you can type it on the screen. If you are going to type it on the screen, click the blank after 'Sample Identifier', type in the identifier with the pop-up soft keyboard like this , and press

to finish or press

to cancel.

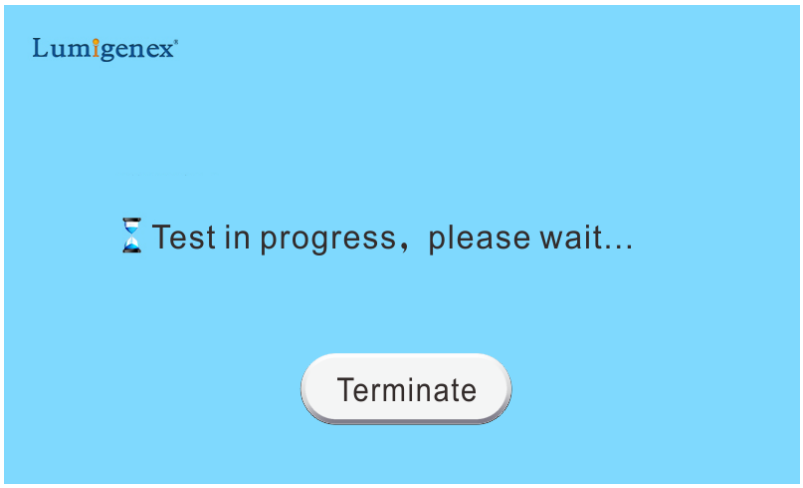
Insert the test strip to the test strip entrance (with the test side upward). If the incubation had been done appropriately before the test, click

, if not, click 

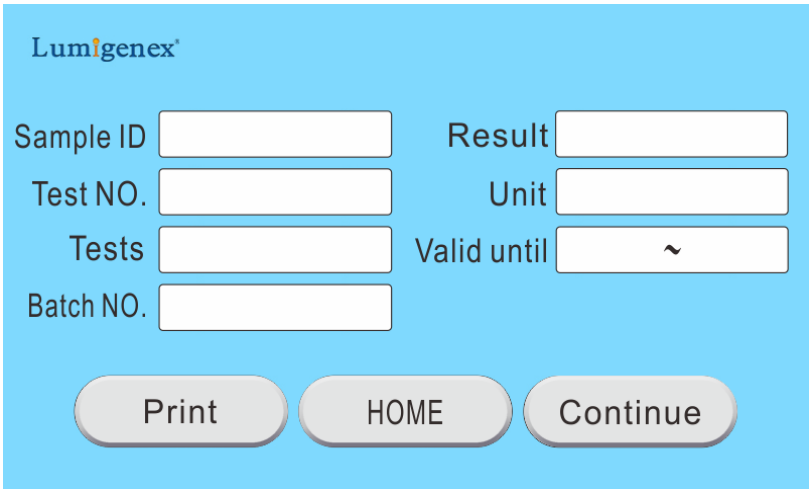
.

If you are going to run the immediate test, please make sure the incubation is completed . If you are going to run the incubation test, please do it as soon as the sample is loaded.

Otherwise could cause a wrong test result.



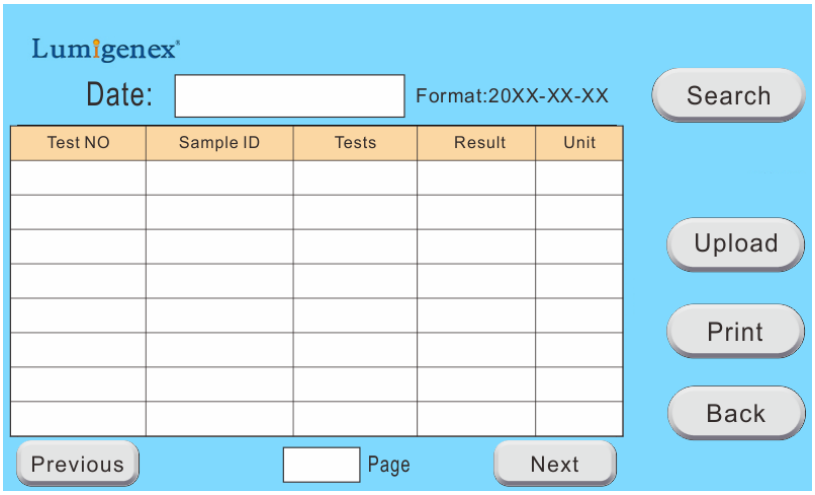
Test result will be shown as the interface below:



Press Print to print out the result or upload it to the LIS system. Click Continue to continue the test, or HOME to end the test.

Displaying

Click Archive on the main interface to enter the Review Interface.



The most recent test record will be shown in this interface as you entered.

7. Control testing

The Time-resolved Fluorescence Immunoanalyzer automatically reads the lot-specific calibration data from the IC card, eliminating the need for calibration by the user. Ensure the lot number of test kit matches the IC card inserted in the instrument.

The control solutions are not included in the product. Please contact us to obtain the control solutions if necessary.

8. Review results

Reviewing test results

Test record can be inquired with date. Click the blank Date: and type in the date for searching. You should type the date with this format: Year-Month-Date. (For example, March 1st, 2015 should be typed as 2015-03-01). Click Search and the information will be shown. You can print out the information you selected or upload them to a PC.

Quality control results

Test record can be inquired with date. Click the blank Date: and type in the date for searching. You should type the date with this format: Year-Month-Date. (For example, March 1st, 2015 should be typed as 2015-03-01). Click Search and the

information will be shown. You can print out the information you selected or upload them to a PC.

Instrument quality control results

Test record can be inquired with date. Click the blank Date: and type in the date for searching. You should type the date with this format: Year-Month-Date. (For example, March 1st, 2015 should be typed as 2015-03-01). Click Search and the information will be shown. You can print out the information you selected or upload them to a PC.

9. Cleaning and Disinfecting the Immunoanalyzer

Recommended cleaning/disinfecting solutions

If the instrument is polluted, it could cause infection to the operator. It is necessary to clean the shell of LTRIC-600 weekly. Make sure you wear gloves and the instrument is disconnected with power during the clean work. Wipe the shell surface, especially the test strip entrance.

Follow the procedures below to clean/disinfect the instrument. Failure to follow these procedures may cause malfunction of the instrument.

- Make sure the instrument is powered off. Unplug the power supply.
- Use only ordinary lint-free cotton swabs/buds and lint-free tissues.
- Ensure that cloth or swabs/buds is only damp, not wet , to protect against moisture entering the instrument.
- DO NOT spray anything onto the instrument and DO NOT immerse it in liquid.

Recommended cleaning/disinfecting solutions

- Mild soapy water
- 75% alcohol

Leave it to dry after cleaning before use.

Daily Maintenance

In order to use LTRIC-600 more safe and more efficient, maintenance work is necessary.

start Maintenance Work

Please start the maintenance work according to [Section 4 Prepare the Test—2 Starting LTRIC-600]

Finish Maintenance Work

Please finish the maintenance work according to [Section 6 Shut Down—1 Shut down]

LTRIC-600 needs to be shut down after the test. Please shut down the machine with following steps:

A. Click the button on the top right corner of the interface to save the data.

B. Turn off the power switch.

'OFF' refers to shut down and 'ON' refers to connected.

Please pull out the plug of the power adapter after you turn off the power switch.

Please keep the power adapter plug disconnect with the socket when LTRIC-600 is unused in order to protect the adapter.

Calibration should be done if the batch of objects to be tested changes.

Pay attention to the valid date of calibration of each object. Pay attention to the batch of immune reagent and calibrate timely.

Any parts or accessories but the print paper of LTRIC-600 has to be maintained or replaced by our company or the agency of our company. Please do not disassemble or replace them by yourself. We are no liable for any mishandling.

Please keep the kit flat during the test. Make sure you wear gloves when you touch the kit. Do not place the kit on the instrument, in case it slips and the sample leak out to pollute the instrument.

Replenish the Print Paper

Print paper is easily to run out. Pay attention to the remainder print paper and replenish it timely. Replenishment should be done according to the requirements of printer installation.

Cleaning the test strip guide

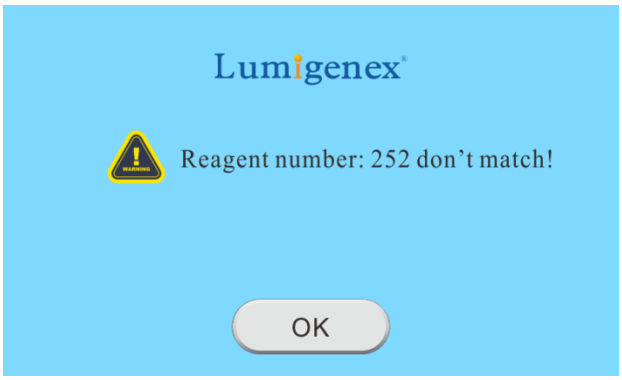
Cleaning the test strip guide daily.

10. Troubleshooting

Errors and unusual behavior without error messages:

1) Button Function Error
2) Crash
3) Abnormal Motor Location
4) IC Card Reading Failure

5) Kit Sample Information Acquiring Failure
6) Data Printing Failure
7) Abnormal Data Printing
8) Invalid or Abnormal Data Process
9) Abnormal or NO Data Displaying
10) Login Failure
11) Wrong Identifier of Items or of Reagents



Faults listed above can be fixed by following steps:

A) Return to the main interface, click SHUT DOWN to shut down the machine, turn off the power switch. Then turn on it again to restart the machine.

B) When the process is not available or it is urgent, you can turn OFF and ON the power switch direct to restart the system.

C) If the IC card can not be uploaded, please pull out and reinsert the card and try again. If the problem still exists, please try again after restarting the machine.

D) If the Test items or reagents identifier doesn't work correctly, please reinsert the test strip and run the test again. If the problem still exists, please try again after restarting the machine.

If the problem still exists after you taking the steps above, please connect our sale representatives or custom service center. We will send you professionals to fix it.

Please do not open this instrument by yourself. Our company is not liable for any mishandling.

11. Specifications

Technical Data

Principle of Detection	Time-Resolved Fluorescence Immunoassay
Process Mode	Item by Item; Batch
Process Ability	<20S/Item

Performance

Sensibility	Signal Value of MDA>40
Accuracy	Relative Deviation -10%≤d≤10%
Precision	CV ≤ 5%
Linearity	Correlation r≥0.990
Output Device	USB, Printer
Dimensions	315 (W) ×255 (D) ×150 (H) mm
Weight	2.4kg
Input Power	72VA
Fuse Capacity	5A

Environmental Set-Up

Temperature	5℃～40℃
Humidity	20%～80%
Atmospheric Pressure	70kPa~106kPa
Power Supply	AC100～240V、72VA 、50/60Hz
Storage	-10℃～55℃,Free from Corrosive gas, Be Ventilative, Indoor

Applied Code

Product Standard	EN /IEC 61010-1:2010, IEC 61010-2-101:2015 ; EN 61326-1:2006, EN 61326-2-6:2006
Overvoltage Category	Category II
Pollution Level	Level 2

12. Warranty

All products have passed the strict quality inspection before packing. If any fault occurs when you use it, our company will fix it according to the Warranty.

Warranty Stipulation

Warranty Period: Within 1 year after installation (installation time will be recorded on the warranty card)

Warranty Coverage: Despite of the correct operation according to the instruction, if fault occurs when still under warranty due to the manufacture issue, our company will fix it for free.

Unwarranted: Even if within warranty period, if it falls into situations as listed below, our company will offer PAID repairment.

1. **Faults or damages are caused by inappropriate operations**
2. **Faults or damages are caused by mis-reforming the accessories or circuitry of this product.**
3. **Faults or damages are caused by inappropriate transportion after installation**
4. **Faults or damages are caused by the strong impact from falling etc.**
5. **Faults or damages are caused by natural disaster such as fire and earthquake etc.**
6. **Faults or damages are caused by abnormal voltage or the use of non-designated power supply.**
7. **Faults or damages that are not caused by the device itself.**
8. **Faults or damages due to repairment, adjustment and reformation by anyone or any institution or the service center not authorized by our company.**

※Consumable accessories are not under warranty.
※Warranty period of renewed accessories will be recorded on the warranty card.
※Repairment over warranty period is a PAID service.

13. Symbols

	Consult instructions for use		Attention, see Instructions for use
IVD	For in vitro diagnostic use only	LOT	Batch Code/ Lot number
REF	Catalog number		Manufacturer
	Store at/Temperature limitation		Do not use if package damaged
	Do not reuse		Use by/ Expiration date
EC REP	Authorized Representative		This way up
	Keep away from sunlight		Handle with care
	Keep dry		Limit stack layers
	BIOLOGICAL RISKS		**Direct current**

14. Contact Lumigenex

Tel: 86-512-80988088

E-mail: info@lumigenex.com

Website: www.lumigenex.com

Lumigenex (Suzhou) Co., Ltd.

Building C24, 218 Xing Hu Street, Suzhou, China 215123

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www.CE-Marking.EU